

Hide menu

Shor's

Video: Introduction to Shor's

1 min

Video: Shor's Algorithm

22 min

Video: Shor Easy Way

5 min

Video: Quantum Fourier Transform

18 min

Video: Quantum Phase Estimation

24 min

Graded Assignment: Assessment 2

Submitted

Your grade: 92%

Your latest: 60% • Your highest: 92% • To pass you need at least 60%. Keep your highest score.

Assessment 2

Next item →

1. What are the functions of Grover's algorithm?

1 / 1 point

☐

To initialize the state of the quantum circuit.

☒

To apply specific transformations based on the value of the unknown function.

☒

Correct

Correct! The Oracle applies transformations to the qubits based on the value of the unknown function.

☐

To perform measurements on qubits.

☒

To represent an unknown function in the computation.

☒

Correct

Correct! The black box (Oracle) is used to represent an unknown function that the algorithm will analyze.

Assignment details

Due Dec 18, 11:59 PM IST

Attempts Unlimited

Retry

2. What is one of the major advantages of Grover's algorithm over classical search algorithms when dealing with an unsorted database?

1 / 1 point

92%

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☒

Grover's algorithm can find an element in $O(\sqrt{N})$ time.

☐

Grover's algorithm works only for sorted databases.

☐

Grover's algorithm uses less memory compared to classical algorithms.

☐

Grover's algorithm can sort the database while searching for an element.

☒

Correct

Correct! Grover's algorithm has a quadratic speedup compared to classical algorithms, which typically require $O(N)$ time for an unsorted database.

3. How does phase change impact the efficiency of Grover's search algorithm?

0 / 1 point

☐

Phase changes help in amplifying the probability of the correct state.

☐

Phase changes are used to reset the quantum state.

☒

Phase changes reduce the overall running time of the algorithm.

☐

Phase changes ensure that the algorithm always finds the correct state.

☒

Incorrect

Not quite. Phase changes affect the probability amplitude, not the time complexity directly.

4. What is the general formula for the Quantum Fourier Transform (QFT) on 1 qubit?

0 / 1 point

☐

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & e^{2\pi i/2} \end{pmatrix}$$

☒

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

☐

$$\begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2} \end{pmatrix}$$

☐

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & e^{2\pi i/2} \end{pmatrix}$$

☒

Incorrect

This is the Hadamard matrix. Review the definition of QFT.

5. Which of the following tools can be used to implement Shor's algorithm?

1 / 1 point

☒

Microsoft Q#

☒

Correct

Correct! Microsoft's Q# is a language specifically designed for quantum computing.

☐

TensorFlow

☐

PyTorch

☒

Qiskit

☒

Correct

Correct! Qiskit is a popular tool for implementing quantum algorithms, including Shor's algorithm.

☒

Cirq

☒

Correct

Correct! Cirq is another tool that can be used to implement Shor's algorithm.